

Tight Confidence Sequences From Universal Portfolio

Based on Orabona and Jun (2024), IEEE Transactions on Information Theory

Xiaoqi (Shirley) Liu

E-Values Reading Group
Department of Statistics, University of Oxford

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Goal: a confidence sequence (CS), i.e., a sequence of confidence intervals $I_t = [\ell_t(X_{1:t}), u_t(X_{1:t})]$ such that

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Main contribution of the paper: tight CSs based on regret bounds for **betting/universal portfolio** algorithms.

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1. Recap and warm-up

- 1.1 Testing by betting
- 1.2 Warm-up with the Krichevsky-Trofimov (KT) betting algorithm

2. Reduction from betting to online portfolio selection

- 2.1 A reformulation using online learning tools
- 2.2 From betting to online portfolio selection

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4. PRECiSE-R70: CS with log-log rate asymptotically in T

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Testing by betting can be generalised through e-values & e-processes (Ramdas and Wang, 2025)

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⇒ Which betting algorithm should we use for building the wealth process?

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Inverting the inequality gives

$$\mathbb{P}\left(\exists t \geq 0 : \left| \mu - \frac{1}{t} \sum_{i=1}^t X_i \right| \geq \sqrt{\frac{2 \ln \frac{2\sqrt{t}}{\delta}}{t}}\right) \leq \delta.$$

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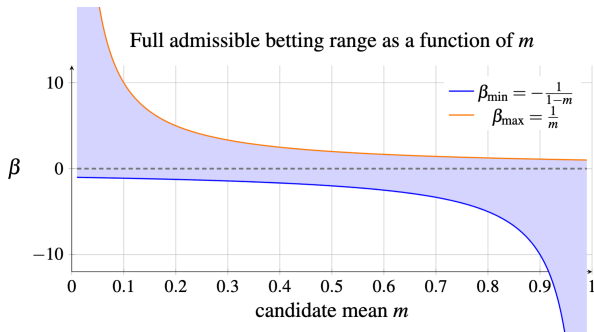
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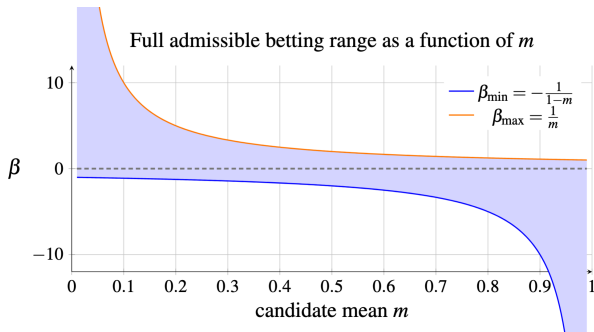


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$\Rightarrow$ **Can we do better than KT?**



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→ Paper focuses on betting algorithms via a **reduction** to **online portfolio selection**.

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Proof sketch.

$$\begin{bmatrix} b \\ 1 - b \end{bmatrix} \cdot \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = b\left(1 + \frac{c}{m}\right) + (1 - b)\left(1 - \frac{c}{1-m}\right) = 1 + c\left(\frac{b}{m} - \frac{1-b}{1-m}\right) = 1 + c\beta.$$

From coin betting to 2-stock online portfolio selection

Lemma 1. Let $m \in [0, 1]$ and $c \in [-m, 1 - m]$. Set two market gains $w_1 = 1 + \frac{c}{m}$ and $w_2 = 1 - \frac{c}{1-m}$ (note $w_1, w_2 \geq 0$ so they are valid market gains). Define $\mathbf{b} = [b, 1 - b]^\top$ to be the play of a 2-stocks portfolio algorithm, where $b \in [0, 1]$. Then, by taking

$$\beta = -\frac{1}{1-m} + \left(\frac{1}{1-m} + \frac{1}{m}\right)b \in \left[-\frac{1}{1-m}, \frac{1}{m}\right]$$

as the signed betting fraction, a continuous coin-betting algorithm on c ensures that the gain in the coin-betting problem **coincides with** the gain in the portfolio selection problem.

Interpretation: Betting on an asymmetric bounded coin **is equivalent to** choosing a portfolio weight in a 2-stock online universal portfolio.

$$\Rightarrow \ln \text{Wealth}_t^{\text{coin}}(\beta_{1:t}, m) = \ln \text{Wealth}_t^{\text{portfolio}}(\mathbf{b}_{1:t}, m), \quad \text{Regret}_t^{\text{coin}}(m) = \text{Regret}_t^{\text{portfolio}}(m)$$

Universal portfolio algorithms

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$$\mathbf{b}_t = \frac{\int_B \mathbf{b} \text{Wealth}_{t-1}(\mathbf{b}) dF(\mathbf{b})}{\int_B \text{Wealth}_{t-1}(\mathbf{b}) dF(\mathbf{b})}, \quad B := \{[b, 1-b]^\top \in \mathbb{R}^2 : b \in [0, 1]\},$$

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with regret guarantee

$$\text{Regret}_T = \max_{\mathbf{b} \in B} \ln \text{Wealth}_T(\mathbf{b}) - \ln \underbrace{\int_B \text{Wealth}_t(\mathbf{b}) dF(\mathbf{b})}_{\text{wealth of the } F\text{-weighted portfolio algorithm}} = o(T).$$

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For the F -weighted universal portfolio algorithm with $F = \text{Beta}(\frac{1}{2}, \frac{1}{2})$, its regret is upper bounded by $\text{Regret}_t \leq \ln \frac{\sqrt{\pi}\Gamma(t+1)}{\Gamma(t+\frac{1}{2})} \leq \frac{1}{2} \ln(4t)$.

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Theorem 1. Let $\delta \in (0, 1)$. Let $X_1, X_2, \dots \in [0, 1]$ such that $\mathbb{E}[X_i \mid X_{1:i-1}] = \mu$ a.s.. Then, PRECiSE-CO96 yields

$$\mathbb{P}\left(\exists t \geq 1 : \max_{\beta \in \left[-\frac{1}{1-\mu}, \frac{1}{\mu}\right]} \sum_{i=1}^t \ln(1 + \beta(X_i - \mu)) \geq \ln \frac{1}{\delta} + \frac{1}{2} \ln(4t)\right) \leq \delta.$$

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Compare with KT guarantee: $\mathbb{P} \left(\exists t \geq 1 : \frac{(\sum_{i=1}^t (X_i - \mu))^2}{2t} \geq \ln \frac{1}{\delta} + \frac{1}{2} \ln(4t) \right) \leq \delta.$

☹ No closed-form concentration, need numerical inversion through bisection

😊 **Never-vacuous** for any sample size

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- Deeper connections to universal coding, Jefferys prior

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$$\mathbb{P}\left(\mu \in \left[\max_{1 \leq i \leq t} \hat{\mu}_i - \varepsilon_i, \min_{1 \leq i \leq t} \hat{\mu}_i + \varepsilon_i\right], \forall t \geq 1\right) \geq 1 - \delta,$$

where for large t , $\varepsilon_t \approx \frac{1}{t} \sqrt{2V_t(\frac{1}{2} \ln t + \ln \frac{1}{\delta})} + \frac{4}{3t}(\frac{1}{2} \ln t + \ln \frac{1}{\delta})$.

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- **Next:** Going from $\frac{1}{t} \sqrt{V_t \ln t}$ (i.e., CLT rate $\times \sqrt{\ln t}$) to $\frac{1}{t} \sqrt{V_t \ln \ln t}$ (i.e., law of the iterated logarithm (LIL) rate, asymptotically optimal)

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By the first-order condition, we have

$$\beta_t^*(m) \in \arg \max_{\beta \in \left[-\frac{1}{1-m}, \frac{1}{m}\right]} \sum_{i=1}^t \ln(1 + \beta(X_i - m)) \approx \frac{\sum_{i=1}^t (X_i - m)}{\sum_{i=1}^t (X_i - m)^2}.$$

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Under $m = \mu$, the LIL gives $\sum_{i=1}^t (X_i - \mu) = \mathcal{O}(\sqrt{V_t \ln \ln t})$, which implies

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$\Rightarrow$ Instead of $F(b) = \text{Beta}(\frac{1}{2}, \frac{1}{2})$, PRECiSE-R70 uses the following $F(\beta)$ from Robbins (1970)

$$F(\beta) \asymp \frac{1}{|\beta| \ln\left(\frac{1}{|\beta|}\right) \left(\ln \ln\left(\frac{1}{|\beta|}\right)\right)^2}.$$

Note $F(\beta)$ is a **proper prior with** $F(\beta) \rightarrow \infty$ as $\beta \rightarrow 0$.

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Theorem 3. Let $\delta \in (0, 1)$. Let $X_1, X_2, \dots \in [0, 1]$ such that $\mathbb{E}[X_i \mid X_{1:i-1}] = \mu$ a.s.. Denote by $\hat{\mu}_t = \frac{1}{t} \sum_{i=1}^t X_i$, $V_t = \sum_{i=1}^t (X_i - \hat{\mu}_t)^2$. Then, PRECiSE-R70 yields a CS such that

$$\mathbb{P}(\mu \in [\ell_t, u_t], \forall t \geq 1) \geq 1 - \delta,$$

where the corresponding confidence width satisfies

$$\max\{u_t - \hat{\mu}_t, \hat{\mu}_t - \ell_t\} \lesssim \frac{1}{t} \sqrt{2V_t(\ln \ln t + \ln \frac{1}{\delta})}.$$

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Compare with

- Empirical Bernstein: $|\hat{\mu}_t - \mu| \lesssim \frac{1}{t} \sqrt{2V_t \ln \frac{1}{\delta}}$ for fixed t
- PRECiSE-CO96: $|\hat{\mu}_t - \mu| \lesssim \frac{1}{t} \sqrt{2V_t(\ln t + \ln \frac{1}{\delta})}$ uniformly in t

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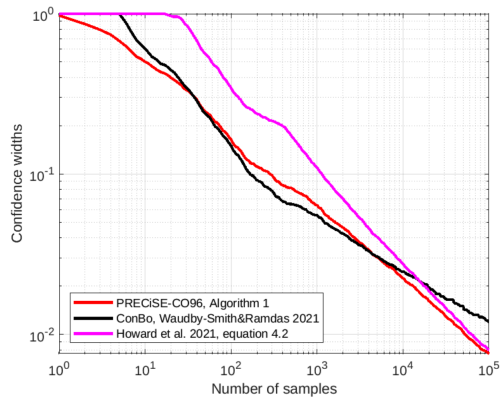
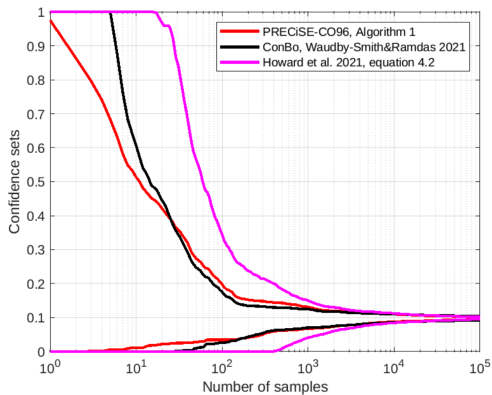


Figure: Time-uniform CSs (left) and their widths (right) for a sequence of **i.i.d. Bernoulli(0.1)** r.v.s.

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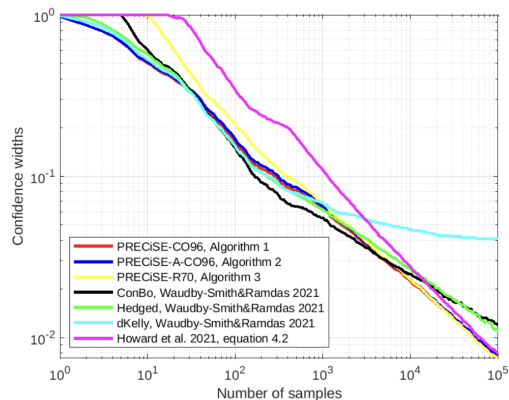
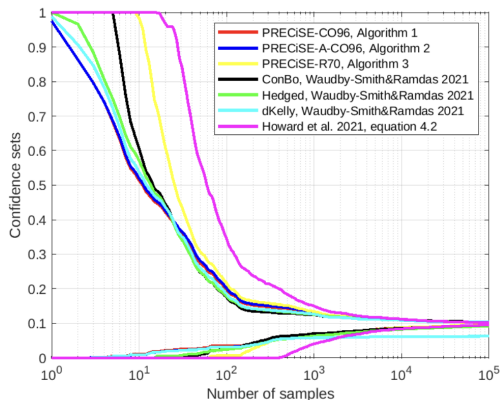


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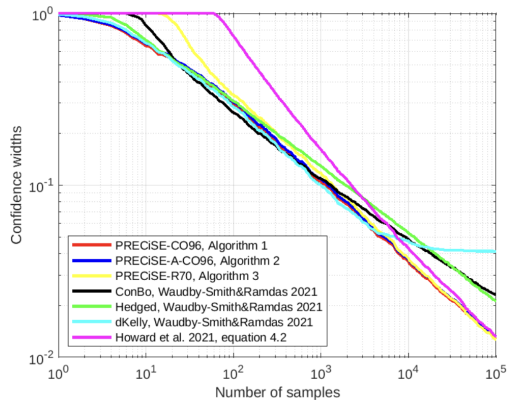
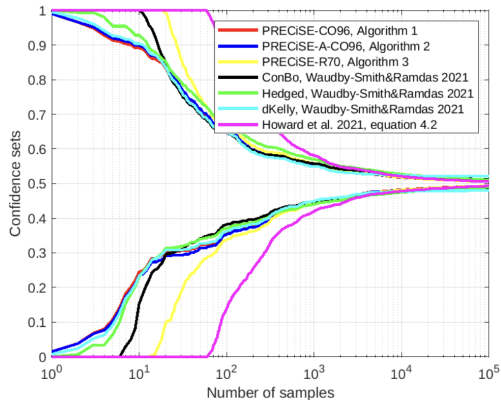


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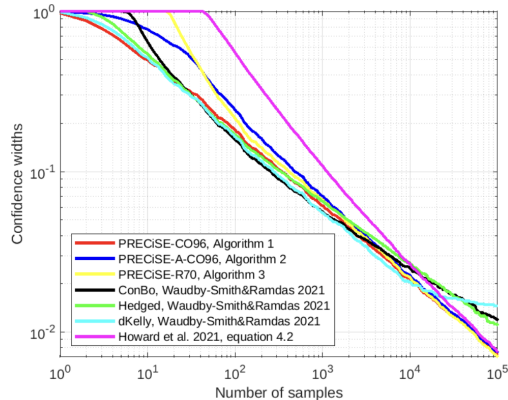
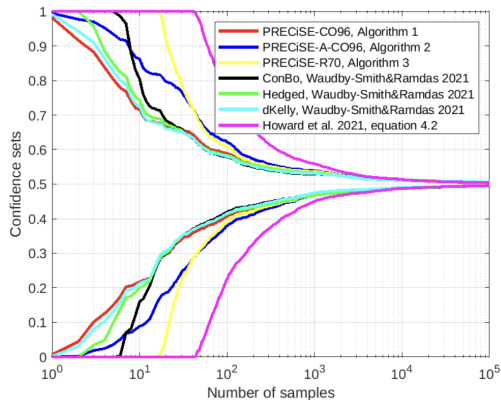


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The paper proves inequalities of the schematic form

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5. **PRECiSE-R70** uses a **zero-centered prior** on the bet $\beta \in [-1, 1]$
 $\Rightarrow$ LIL rate asymptotically

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